Demand-Driven Staffing Optimization

Executive Summary — Joanna Ni

Can staffing to predicted demand, instead of a flat schedule, protect service on busy days without raising labor cost?

16.7% → 11.1%	30 / 30	+0.4%	30.5%
AVG. FORECAST ERROR (MAPE), BASELINE VS. MODEL	STORES WHERE THE MODEL BEAT THE BASELINE	AVG. LABOR COST, DEMAND-DRIVEN VS. FIXED	OF DAYS A FIXED SCHEDULE WOULD UNDERSTAFF

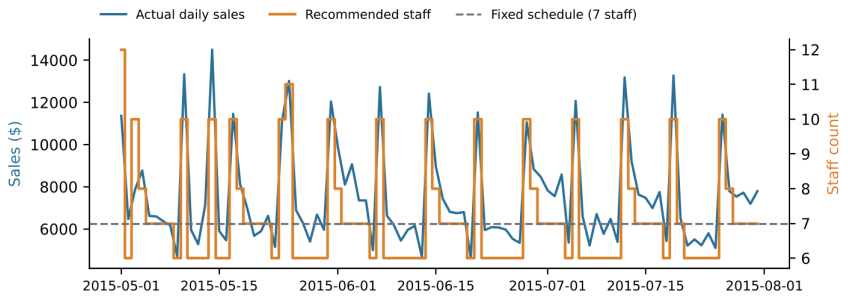
Approach

- Trained a gradient boosting model on Rossmann Store Sales data (Kaggle; 1,115 stores, daily, 2013–2015) using day of week, promotions, and holiday flags. Tested on the final 3 months, which the model never saw.
- Converted each day's forecast into a staffing level with a documented assumption (2 minimum staff + 1 per $1,500 in predicted sales; $15/hr, 8-hour shifts). Real staffing data is not public.
- Compared against a flat schedule sized to average demand. Developed on one store (Store 85), then repeated on a random sample of 30 stores.

Key findings

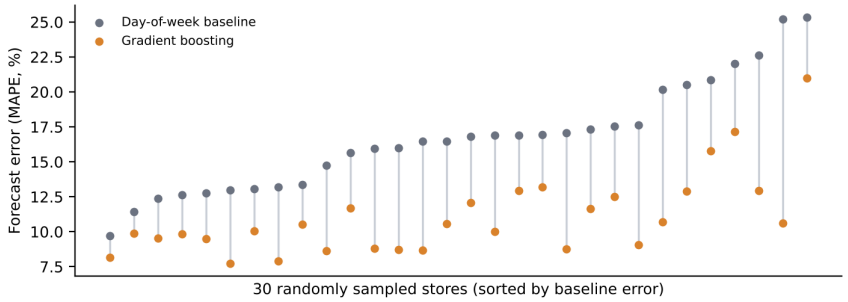
- **The forecast improves in every store tested.** Average error fell by about one-third (16.7% → 11.1% MAPE), and the model beat the day-of-week baseline in all 30 stores.
- **Cost is roughly neutral; risk is not.** Demand-driven staffing cost +0.4% on average (cheaper in 14 of 30 stores), while a flat schedule would be understaffed on 30.5% of days.
- **Store 85 deep dive:** model error 13.0% → 8.7%; demand-driven staffing cost $2,160 more (+2.8%) over 92 days, but a 7-person flat schedule would be short on 23 of 92 days, all high-demand days (mostly Sundays and Mondays).

Store 85: demand vs. staffing recommendation (test period)



The flat schedule (gray) cannot follow the weekly demand spikes; the recommended staffing (orange) does.

All 30 sampled stores: forecast error, baseline vs. model



Each pair is one store; the model (orange) sits below the baseline (gray) in every store.

Limitations: staffing rule and wage are assumptions, not real schedules; one 3-month test window; features limited to calendar and promotion signals.
Code and data: github.com/jcannani/restaurant-staffing-optimization | Tools: Python (pandas, scikit-learn, matplotlib) | nijoanna16@gmail.com